

An Active Learning Plugin in napari to Fine-Tune Models for Large-scale Bioimage Analysis

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Outline

- Motivation
 - Related work
- Methods
- Results
 - Implementation details
 - Use case
- Concluding remarks

Motivation

Motivation

Adoption of deep learning methods for bioimage analysis has grown exponentially in recent years ...

Fine-tuning models (transfer learning)

General computer vision databases:

- ImageNet
- Segment Anything 1 Billion (SA-1B)

De-novo training

Bioimage analysis databases:

- LIVECell
- TissueNet
- CellSeg
- ...

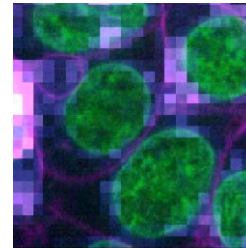
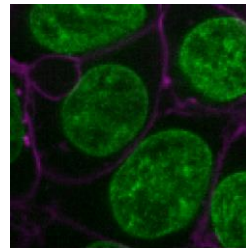
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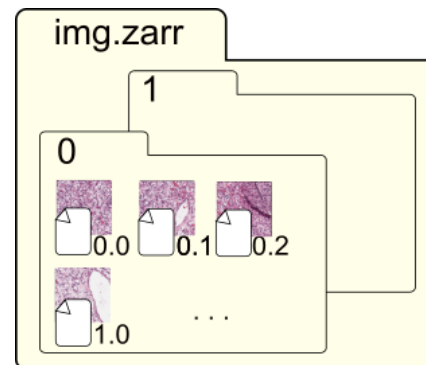
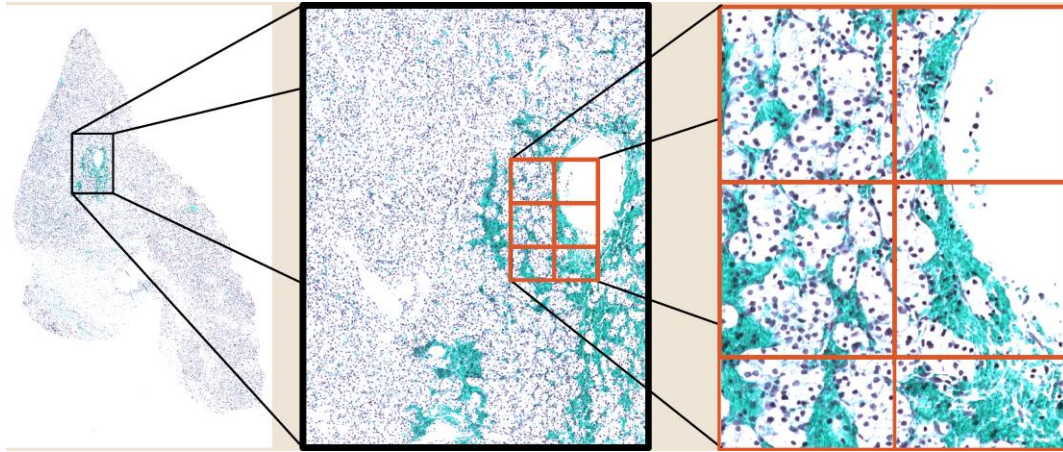
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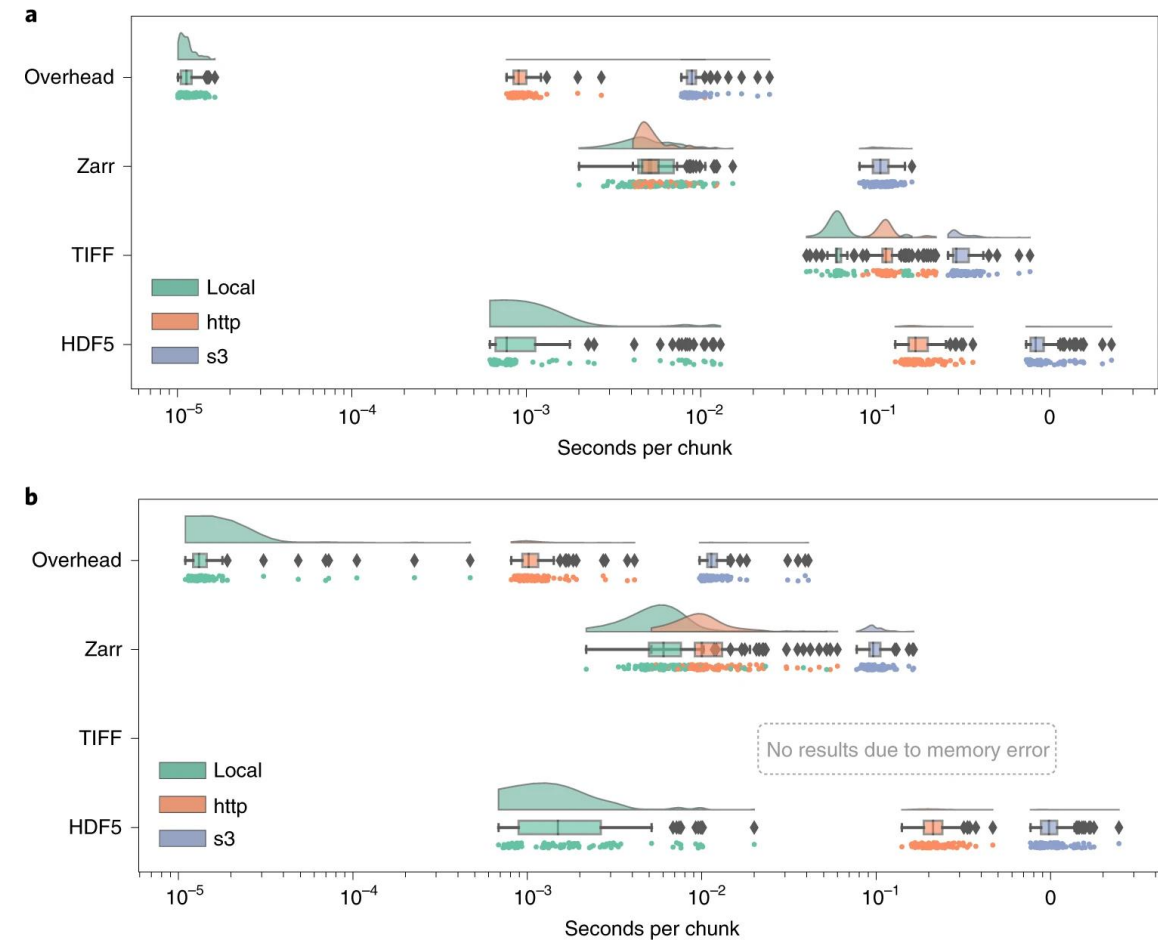
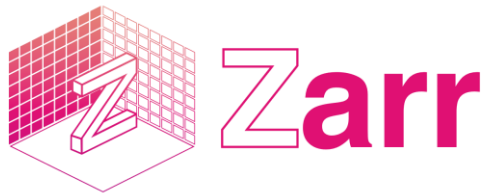
Motivation

Next Generation File Formats (OME-Zarr)



Motivation

Next Generation File Formats (OME-Zarr)



Motivation

Next Generation File Formats (OME-Zarr)

Image processing

- Python (zarr-python)
- Fiji (MOBYE plugin)
- QuPath



Visualization tools

- napari
- neuroglancer
- Viv
- webKnossos



Motivation

[Use case] Segmentation of structures in biological image data

Deep learning segmentation methods:

- Cellpose
- Stardist
- Micro-SAM

Motivation

[Use case] Segmentation of structures in biological image data

Training deep learning models for bioimage analysis involves:

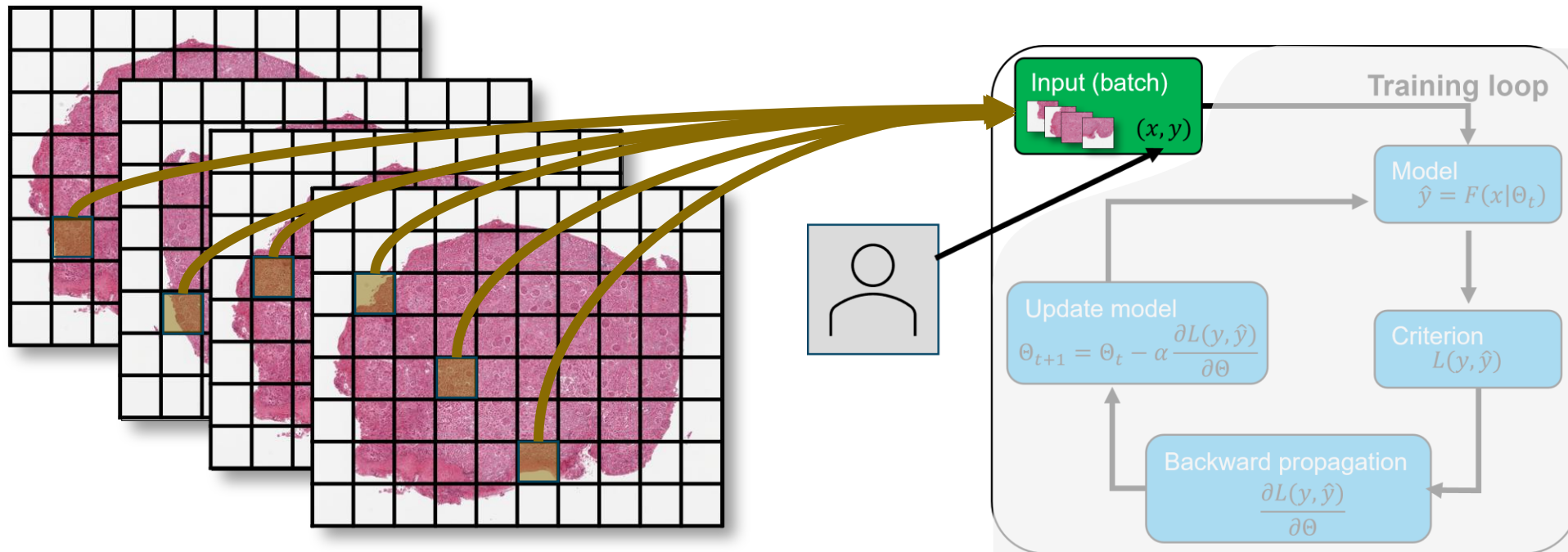
- extracting image tiles
 - annotating each of those tiles
 - storing pairs of tiles and annotations
- Shareable?
 - Reproducible?

Methods

Methods

Active Learning

Studies methods for human-in-the-loop workflows that avoid overwhelming the annotator

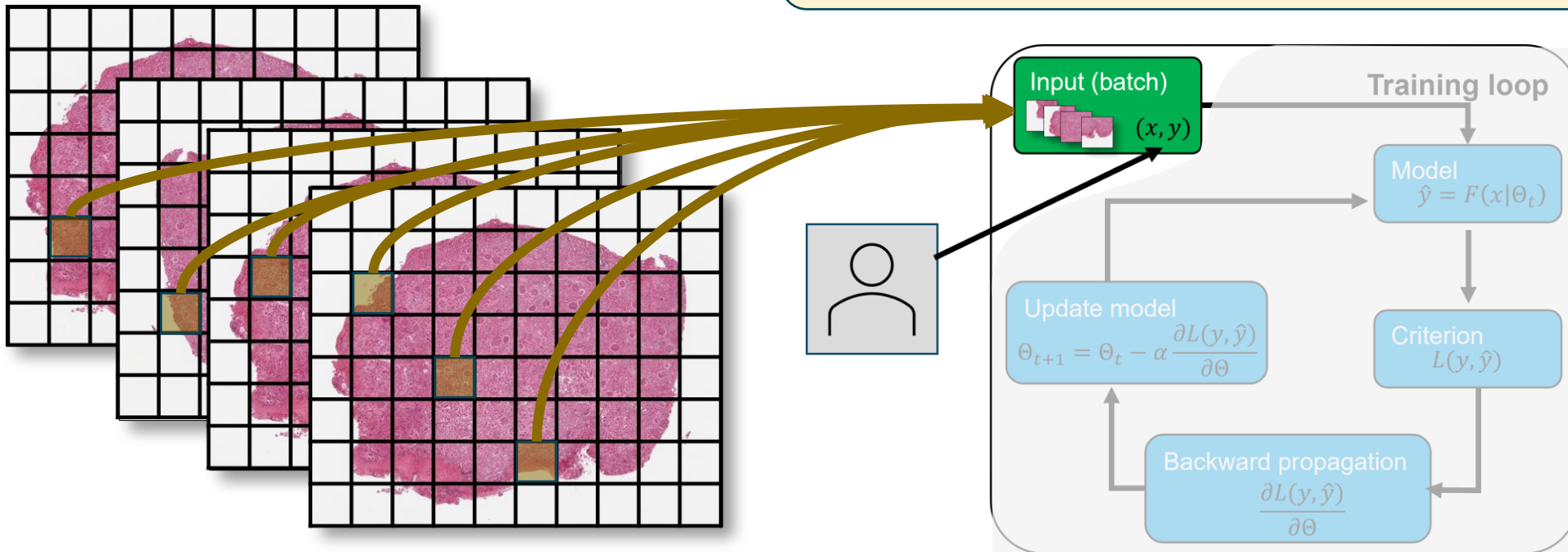


Methods

Active Learning

Acquisition Functions

What is the next sample that the annotator should label?



Methods

Active Learning

Acquisition Functions –Bayesian Active Learning by Disagreement (BALD)

$$\mathbb{I}[y, \omega | \mathbf{x}, \mathcal{D}_{\text{train}}] = \mathbb{H}[y | \mathbf{x}, \mathcal{D}_{\text{train}}] - \mathbb{E}_{p(\omega | \mathcal{D}_{\text{train}})} [\mathbb{H}[y | \mathbf{x}, \omega]]$$

How much can we reduce the model uncertainty when predicting y using the training dataset.

Methods

Active Learning

Acquisition Functions –Bayesian Active Learning by Disagreement (BALD)

$$\mathbb{I}[y, \omega | \mathbf{x}, \mathcal{D}_{\text{train}}] = \mathbb{H}[y | \mathbf{x}, \mathcal{D}_{\text{train}}] - \mathbb{E}_{p(\omega | \mathcal{D}_{\text{train}})} [\mathbb{H}[y | \mathbf{x}, \omega]]$$

How much can we reduce the model uncertainty when predicting y using the training dataset.

Following [Gal et al. 2017]:

$$\begin{aligned} \mathbb{I}[y, \omega | \mathbf{x}, \mathcal{D}_{\text{train}}] &\approx - \sum_c \left(\frac{1}{T} \sum_t \hat{p}_c^t \right) \log \left(\frac{1}{T} \sum_t \hat{p}_c^t \right) \\ &\quad + \frac{1}{T} \sum_{c,t} \hat{p}_c^t \log \hat{p}_c^t =: \hat{\mathbb{I}}[y, \omega | \mathbf{x}, \mathcal{D}_{\text{train}}] \end{aligned}$$

Methods

Active Learning

Following [Gal et al. 2017]:

Approximate the Bayesian process $\hat{\mathbf{p}}_c^t$ by introducing Dropout into the model at inference time.

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Methods

Active Learning

```
class DropoutEval0verrider(torch.nn.Module):
    def __init__(self, dropout_module):
        ...
        self._dropout = type(dropout_module)(
            dropout_module.p, inplace=dropout_module.inplace)

    def forward(self, input):
        ...
        self._dropout.training = True
        out = self._dropout(input)
        ...
```

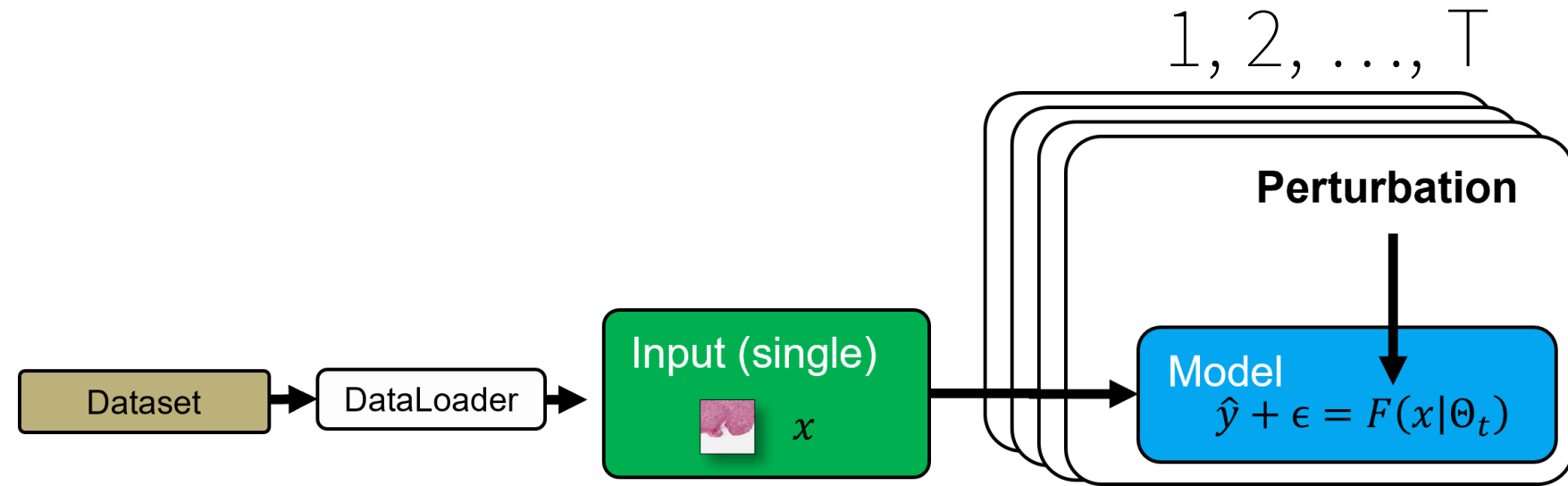
Approximate the Bayesian process $\hat{\mathbf{p}}_c^t$ by introducing Dropout into the model at inference time.

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Methods

Active Learning



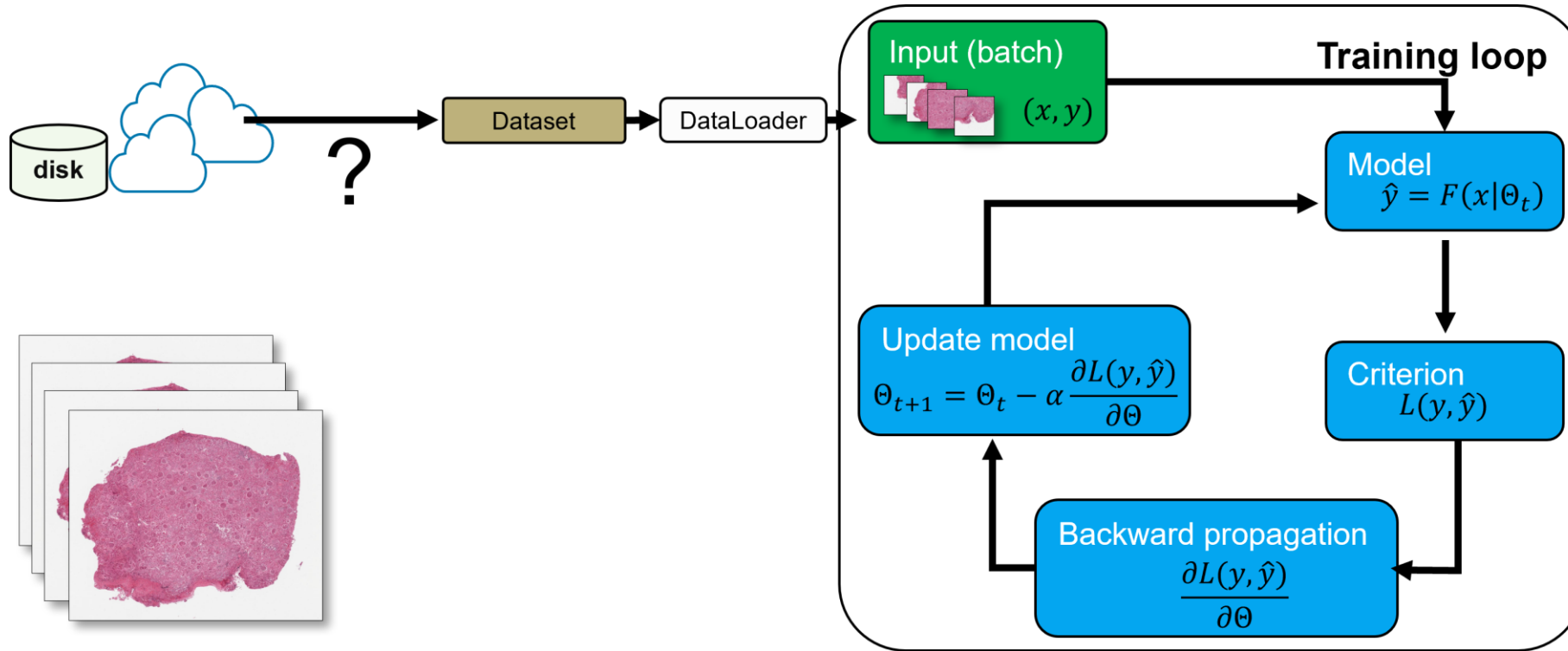
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Methods

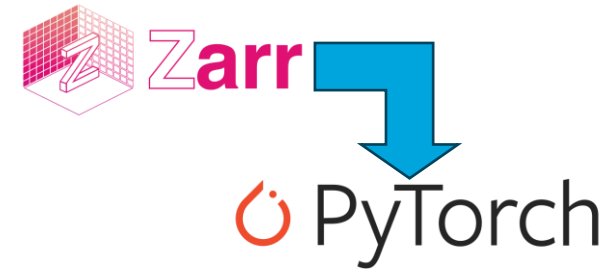
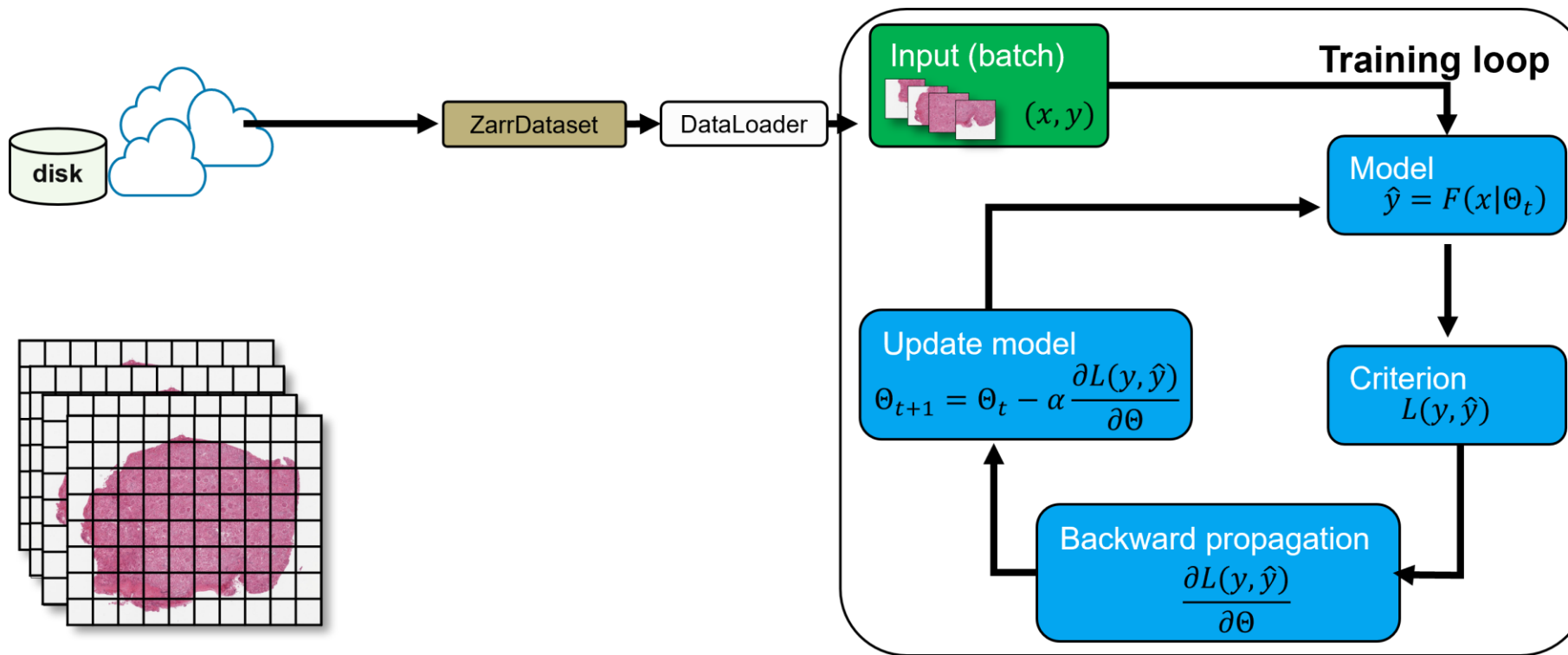
Active Learning



Methods

Active Learning

<https://github.com/TheJacksonLaboratory/zarrdataset>



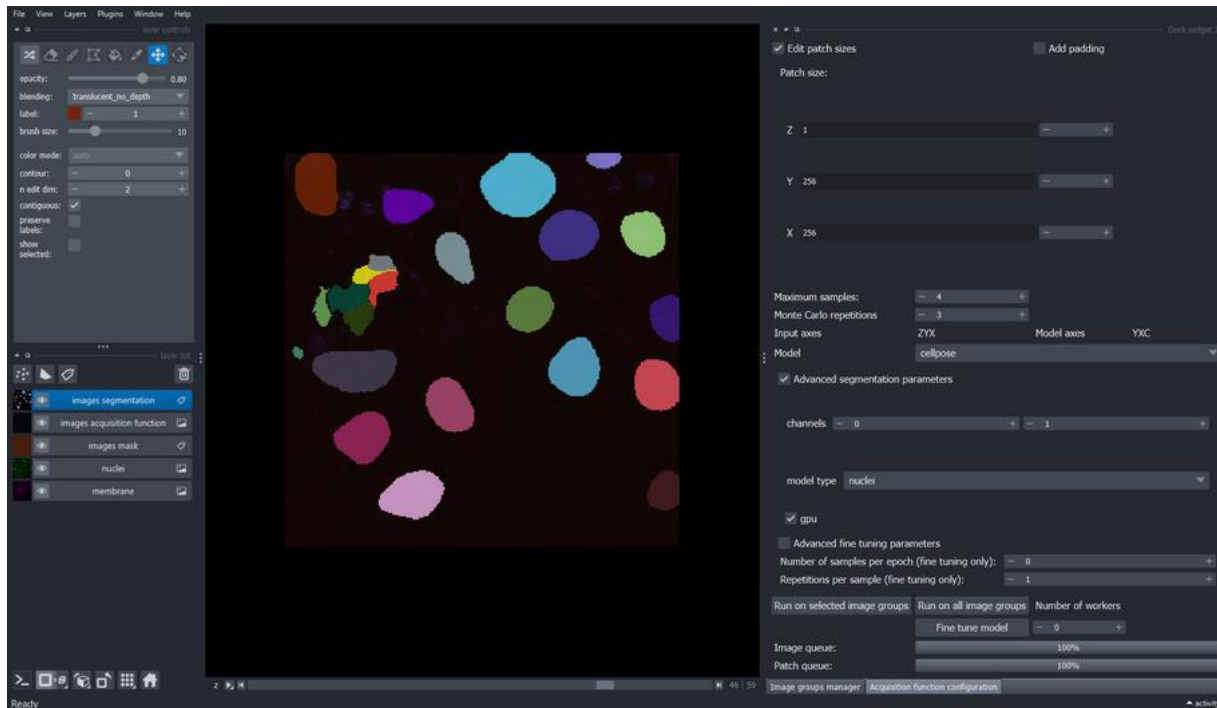
Results

Results

Active learning plugin for napari

<https://github.com/TheJacksonLaboratory/activelearning>

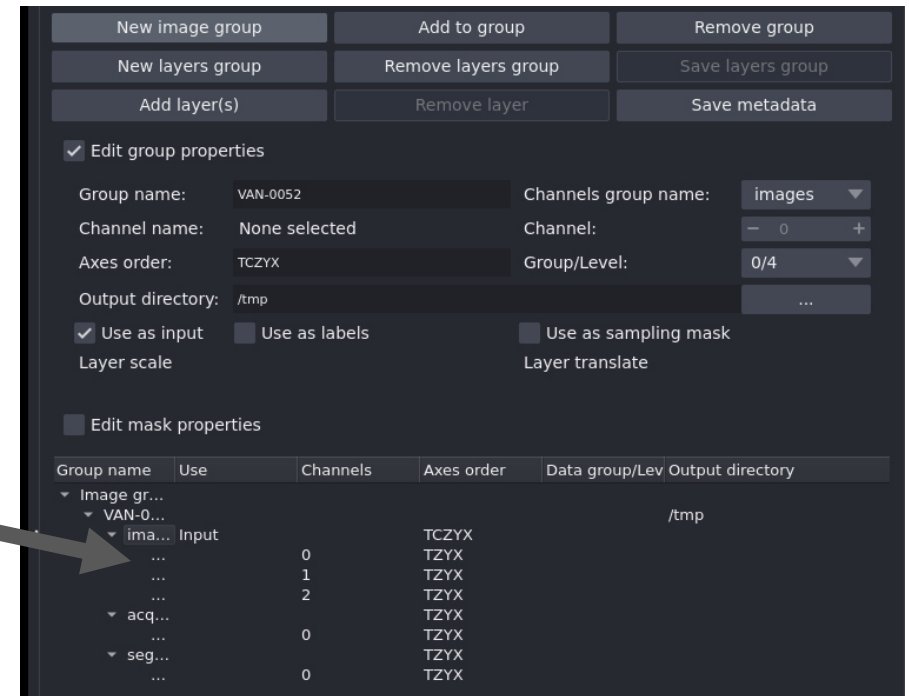
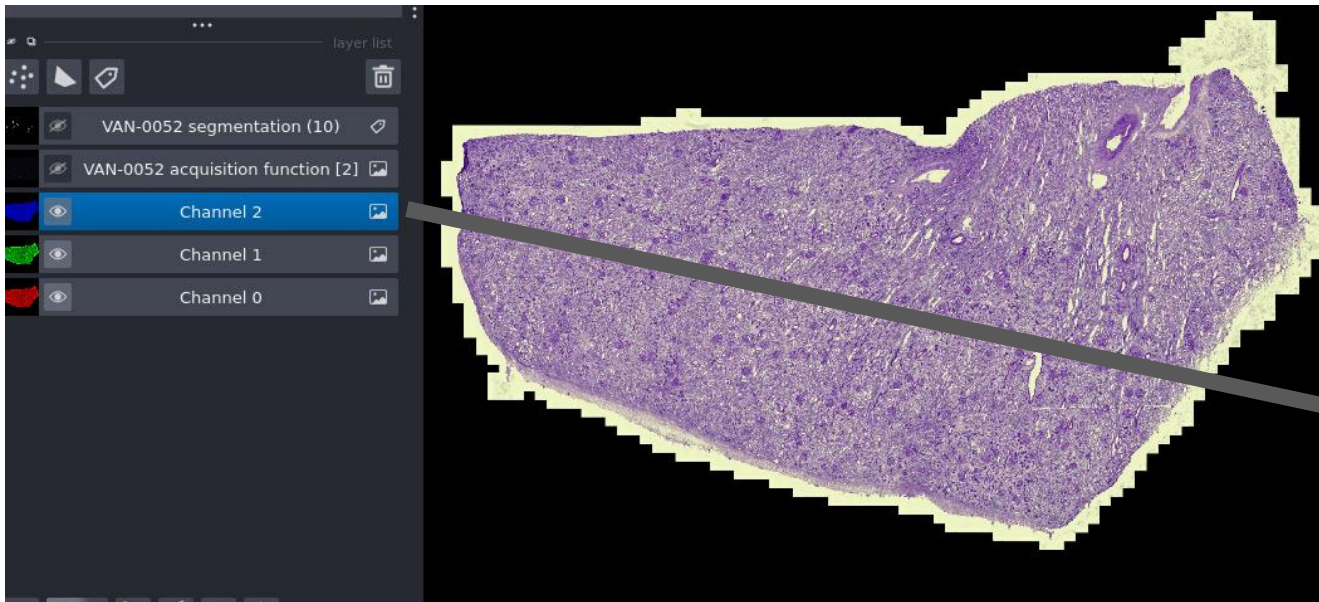
<https://pypi.org/project/napari-activelearning>



Results

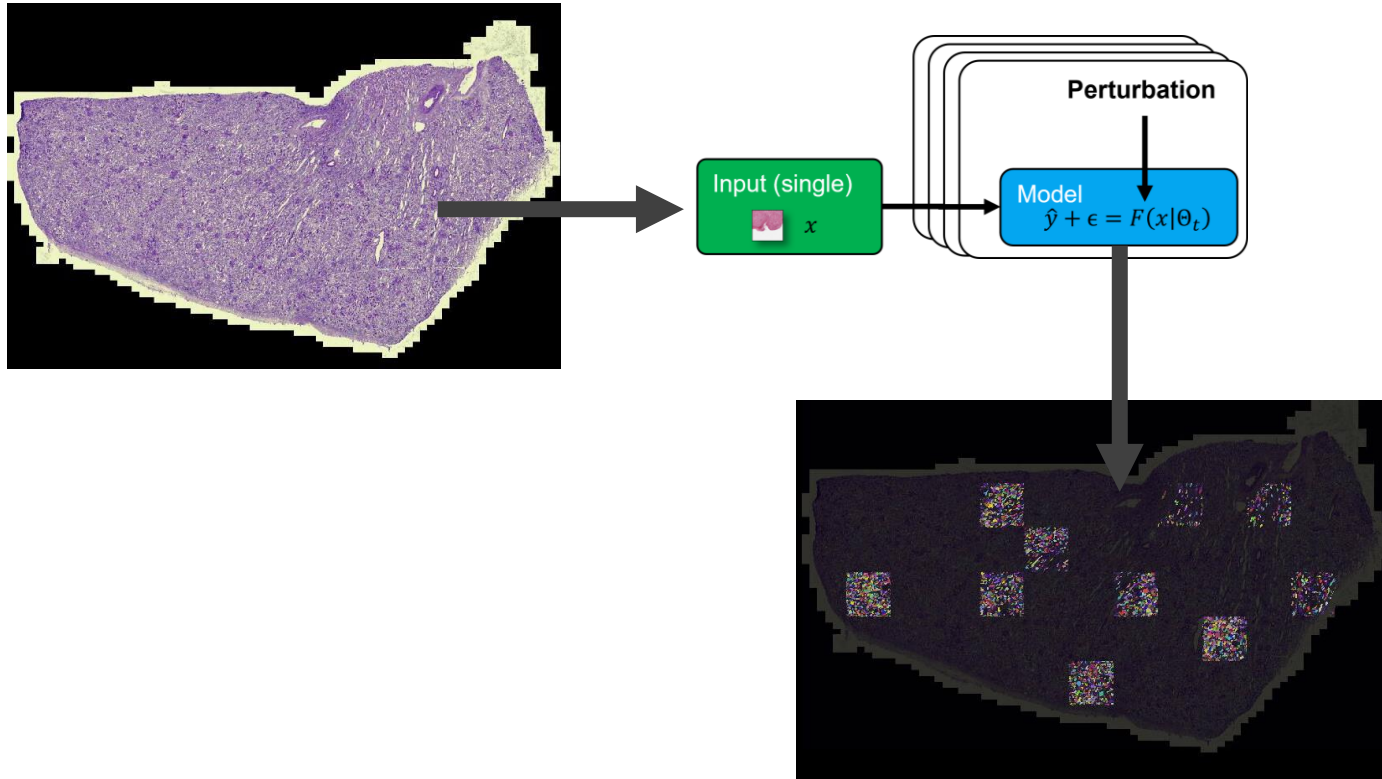
Active learning plugin
for napari

Image Groups Manager



Results

Active learning plugin
for napari

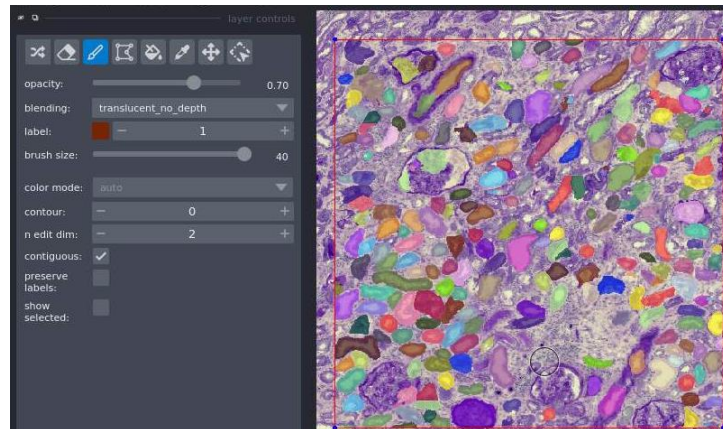
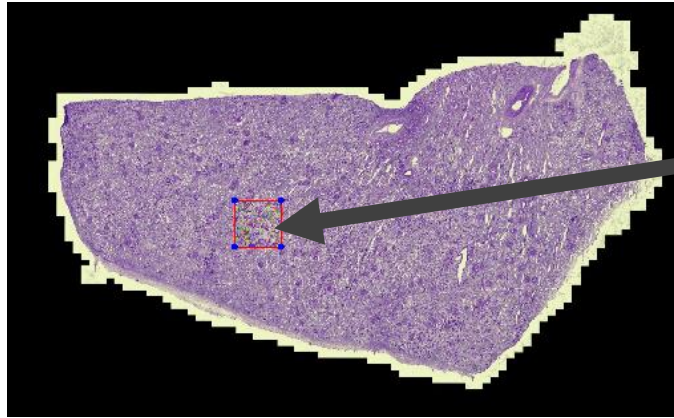


Acquisition Function Manager

The screenshot shows the "Acquisition Function Manager" interface. It features a dark background with various settings and controls. At the top, there are checkboxes for "Edit patch sizes" (checked) and "Add padding". Below this, the "Patch size:" section contains four rows of sliders for T, Z, Y, and X axes, each with a minus and plus button. The Y and X axes are set to 8192. The "Maximum samples:" section has sliders for "Maximum samples" (set to 10) and "Monte Carlo repetitions" (set to 3). The "Input axes" are set to "TZYX" and "Model axes" to "YXC". The "Model" is set to "cellpose". The "Advanced segmentation parameters" section includes a "channels" slider (set to 0) and a "model type" dropdown (set to "cyto3"). The "Advanced fine tuning parameters" section includes a "gpu" checkbox (checked), a "Number of samples per epoch (fine tuning only):" slider (set to 20), and a "Repetitions per sample (fine tuning only):" slider (set to 4). At the bottom, there are buttons for "Run on selected image groups", "Run on all image groups", and "Number of workers" (set to 0). The "Image queue:" and "Patch queue:" are both set to 100%.

Results

Active learning plugin
for napari



Labels Manager

Acquisition value	Sampling centre	Sampling top-left	Sampling bottom-right
Labeled groups			
Labels of group: VAN-0052			
0.58390725	(0.5, 0.5, 23...	(0.0, 2048, ...	(1.1, 2560, 6144)
0.58009577	(0.5, 0.5, 23...	(0.0, 2048, ...	(1.1, 2560, 2560)
0.5714441	(0.5, 0.5, 12...	(0.0, 1024, ...	(1.1, 1536, 2560)
0.5711871	(0.5, 0.5, 23...	(0.0, 2048, ...	(1.1, 2560, 1024)
0.57117146	(0.5, 0.5, 28...	(0.0, 2560, ...	(1.1, 3072, 5120)
0.5708867	(0.5, 0.5, 17...	(0.0, 1536, ...	(1.1, 2048, 3072)
0.5652977	(0.5, 0.5, 23...	(0.0, 2048, ...	(1.1, 2560, 4096)
0.5600364	(0.5, 0.5, 33...	(0.0, 3072, ...	(1.1, 3584, 3584)
0.5576855	(0.5, 0.5, 12...	(0.0, 1024, ...	(1.1, 1536, 4608)
0.5135963	(0.5, 0.5, 12...	(0.0, 1024, ...	(1.1, 1536, 5632)

<<

<

>

>>

Edit current labels

Commit changes

Remove labels

Remove labels group

Results

[Use case] Fine-tuning a Cellpose model

```
class CellposeTunable(TunableMethod):
    model_axes = "YXC"
    _channel_axis = 2

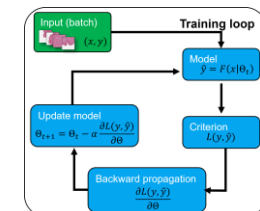
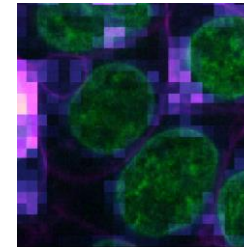
    def __init__(self):
        super().__init__()
        self._model = None
        self._model_dropout = None
        ...

    def _run_pred(self, img, *args, **kwargs):
        # Code to compute probabilities for a single patch image

    def _run_eval(self, img, *args, **kwargs):
        # Code to apply inference on a single patch image

    def _fine_tune(self, train_dataloader, val_dataloader) -> bool:
        # Code wrapping fine-tuning pipeline
```

```
def register_model(model_name: str,
                   model: TunableMethod):
    ...
```



Results

[Use case] Fine-tuning a Cellpose model

Tutorials

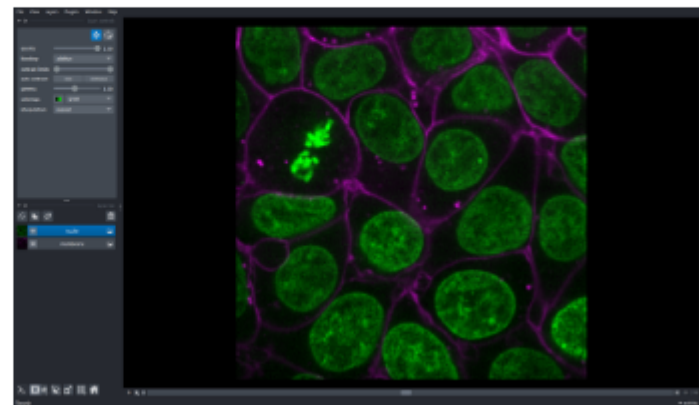
Fernando Cervantes

Tutorial: How to fine tune a Cellpose model with the Active Learning plugin for Napari

CELLPOSE

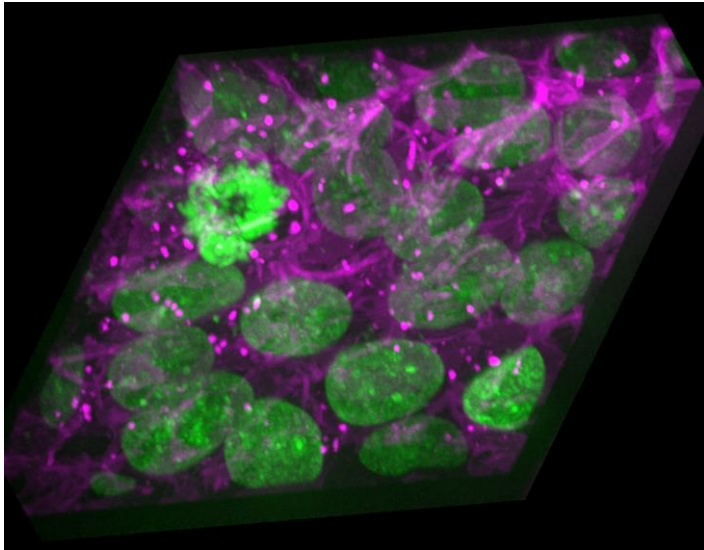
TRANSFER LEARNING

Tutorial for fine-tuning the “nuclei” model from `cellpose` to new data.



Results

[Use case] Fine-tuning a Cellpose model



256x256x60 voxels
Sci-kit Image: 3D fluorescence
microscopy image of cells from the
Allen Institute for Cell Science

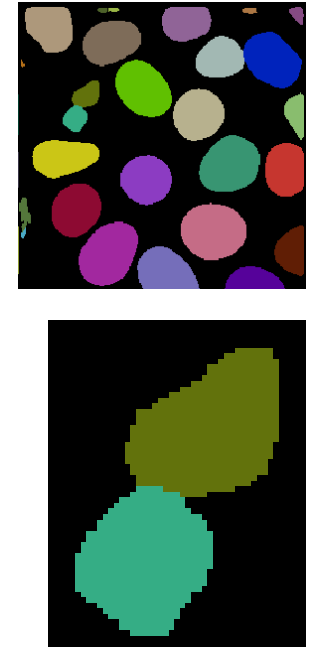
1. Segmentation with
pre-trained
Cellpose's nuclei model



2. Labels correction
in napari

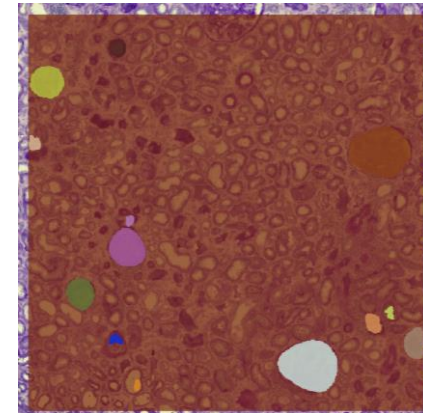
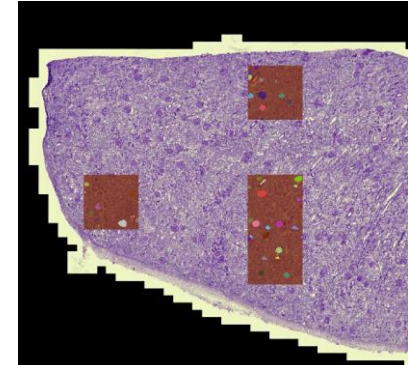
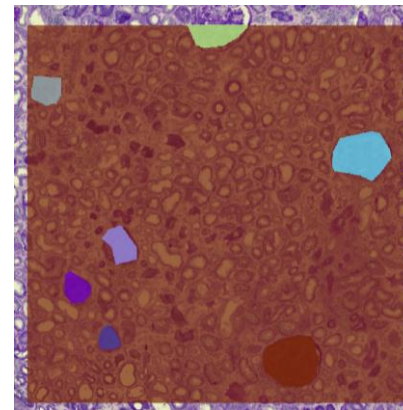
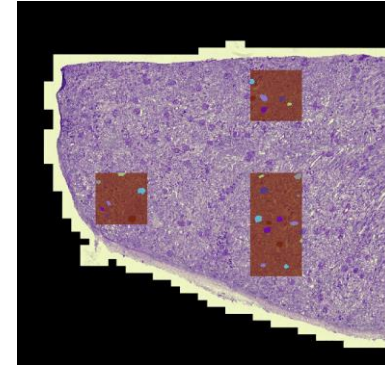
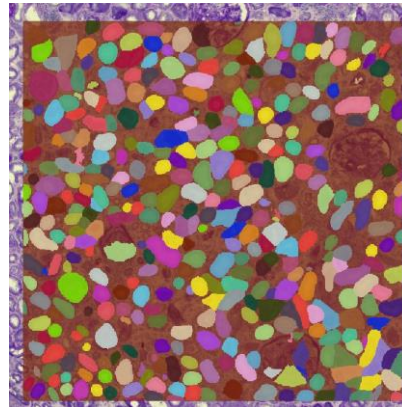
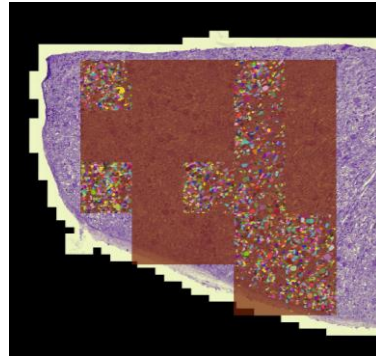
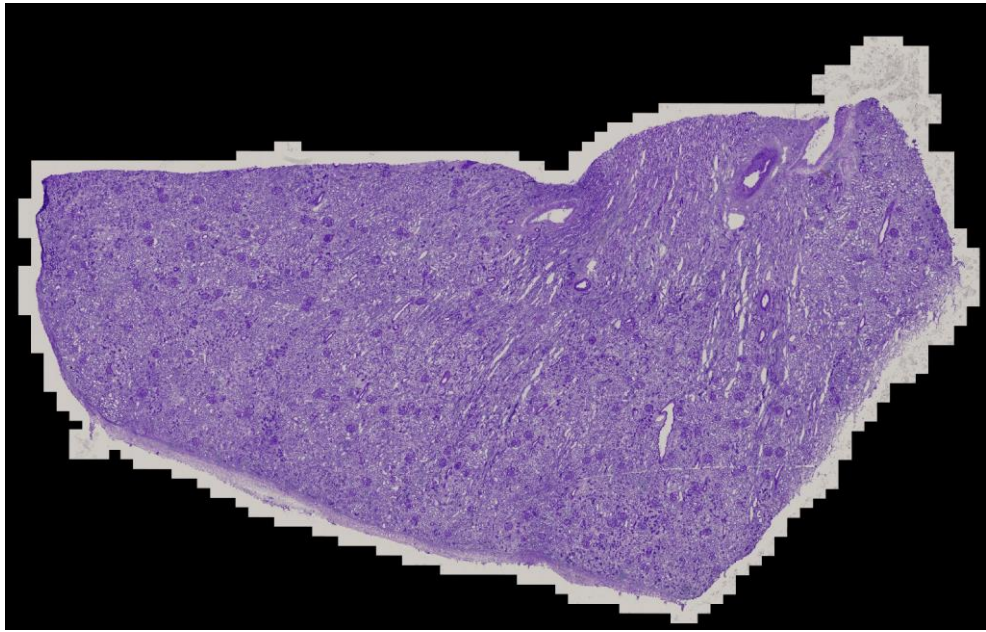


3. Fine-tuning of
Cellpose's nuclei model



Results

[Use case] Fine-tuning a Cellpose model



65,999 x 108,240 pixels (RGB)

HuBMAP - Hacking the Kidney

Jain, Y., Godwin, L.L., Ju, Y. et al. Segmentation of human functional tissue units in support of a Human Reference Atlas.

Commun Biol 6, 717 (2023). <https://doi.org/10.1038/s42003-023-04848-5>.

Concluding remarks

Concluding remarks

- Leveraging NGFF and napari as visualization tool we were able to execute a fine-tuning workflow on large-scale image data with little data preparation
- Not restricted to bioimage data and can be extended to any large-scale image modality
- Finally, the plugin can be used for transfer learning or simply as an interface for DL methods without GUI

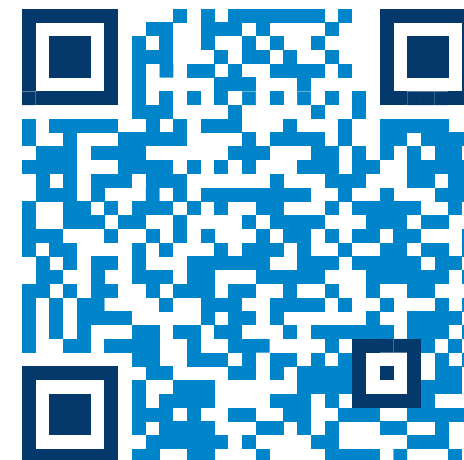


Acknowledgments



<https://github.com/TheJacksonLaboratory/activelearning>

<https://pypi.org/project/napari-activelearning>





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